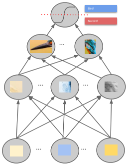
# Deep Learning

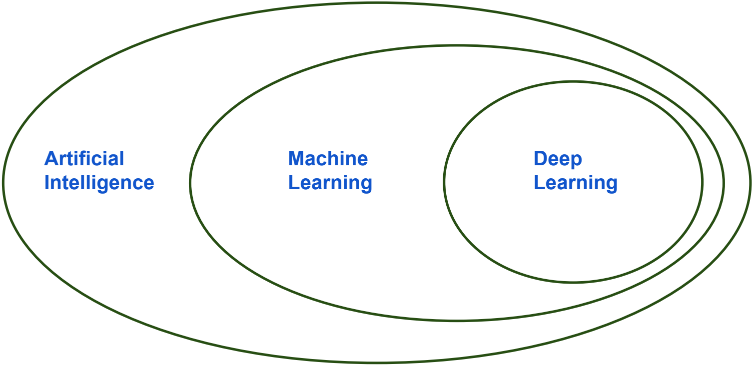
# Introduction




## Learning goals

- Relationship of DL and ML
- Concept of representation or feature learning
- Use-cases and data types for DL methods

# WHAT IS DEEP LEARNING




- Deep learning is a subfield of ML based on artificial neural networks.

# DEEP LEARNING AND NEURAL NETWORKS

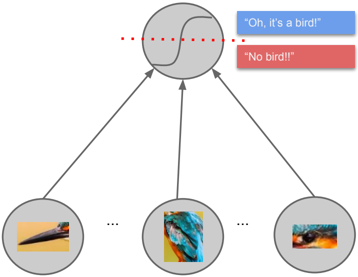
- Deep learning itself is not *new*:
  - Neural networks have been around since the 70s.
  - *Deep* neural networks, i.e., networks with multiple hidden layers, are not much younger.

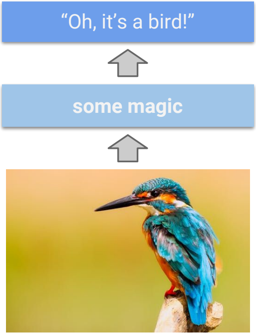
- Why everybody is talking about deep learning now:
  1. Specialized, powerful hardware allows training of huge neural networks to push the state-of-the-art on difficult problems.
  2. Large amount of data is available.
  3. Special network architectures for image/text data.
  4. Better optimization and regularization strategies.

# IMAGE CLASSIFICATION WITH NEURAL NETWORKS

*“Machine learning algorithms, inspired by the brain, based on learning multiple levels of representation/abstraction.”*

Y. Bengio



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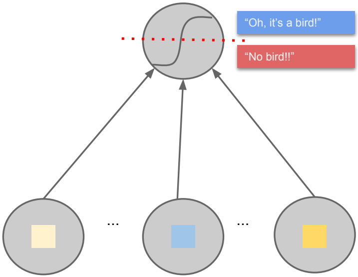


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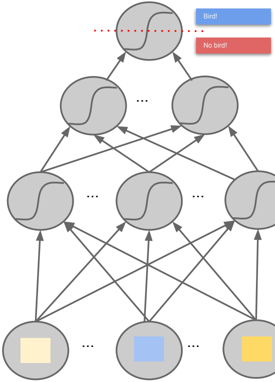


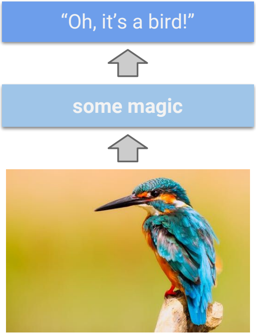

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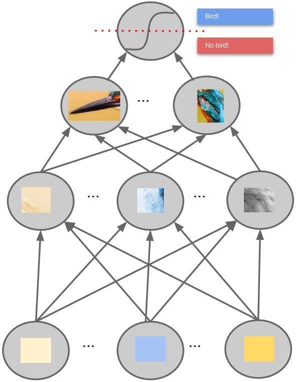
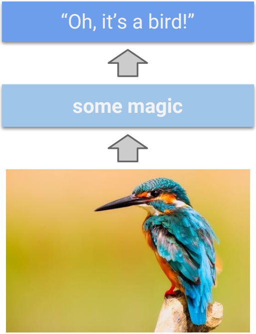


# POSSIBLE USE-CASES

**Deep learning can be extremely valuable if the data has these properties:**

- It is high dimensional.
- Each single feature itself is not very informative but only a combination of them might be.
- There is a large amount of training data.

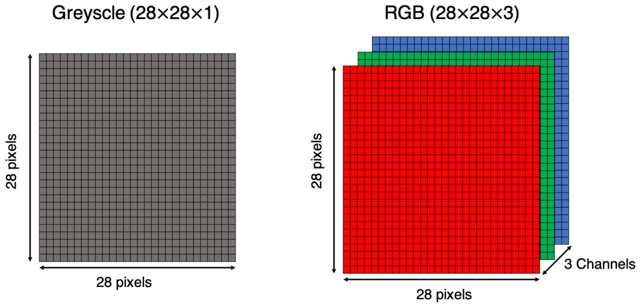
**This implies that for tabular data, deep learning is rarely the correct model choice.**

- Without extensive tuning, models like random forests or gradient boosting will outperform deep learning most of the time.
- One exception is data with categorical features with many levels.

# POSSIBLE USE-CASE: IMAGES

- **High Dimensional**: A color image with $255 \times 255$ (3 Colors) pixels already has 195075 features.
- **Informative**: A single pixel is not meaningful in itself.
- **Training Data**: Depending on applications huge amounts of data are available.

Architecture: **C**onvolutional **N**eural **N**etworks (CNN)



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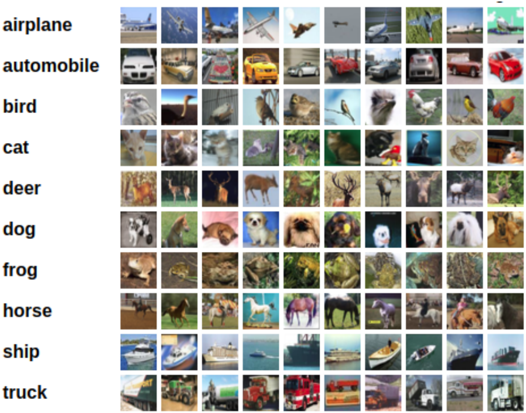



**Figure:** Example for image classification (Krizhevsky, 2009)

**Image classification** tries to predict a single label for each image.
CIFAR-10 is a well-known dataset used for image classification. It consists of $60,000$ $32x32$ color images containing one of 10 object classes, with 6000 images per class.

# POSSIBLE USE-CASE: IMAGES

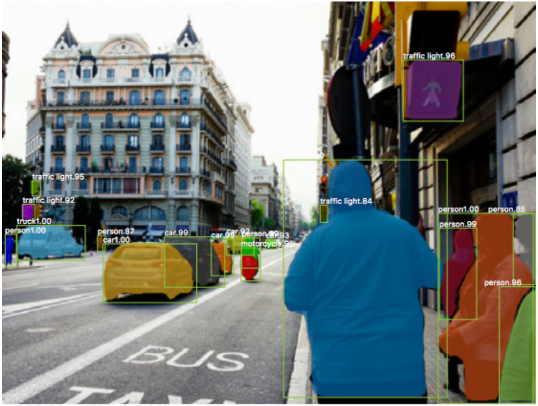



**Figure:** Example for object detection (He et al., 2017)

**Object Detection** Mask R-CNN is a general framework for instance segmentation, that efficiently detects objects in an image while simultaneously generating a high-quality segmentation mask for each instance.

# POSSIBLE USE-CASE: IMAGES

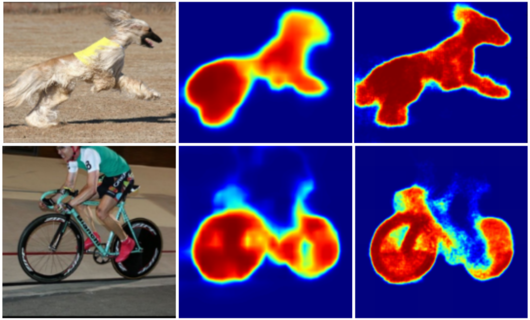


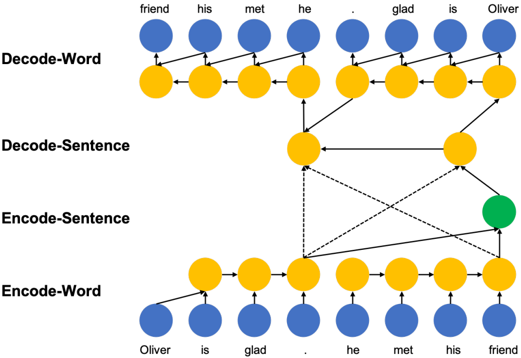
**Figure:** Example for image segmentation (Noh et al., 2015)

**Image segmentation** partitions the image into (multiple) segments.

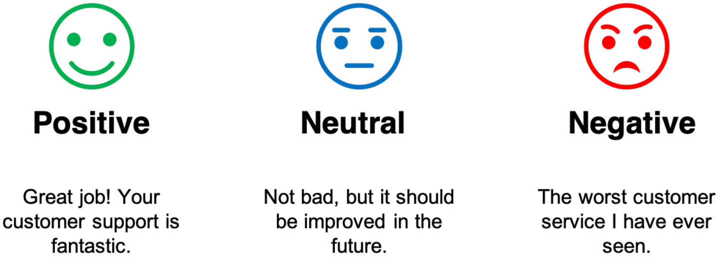
# POSSIBLE USE-CASE: TEXT

- **High Dimensional**: Each word can be a single feature (300000 words in the German language).
- **Informative**: A single word does not provide much context.
- **Training Data**: Huge amounts of text data available.

Architecture: **R**ecurrent **N**eural **N**etworks (RNN)



# POSSIBLE USE-CASE: TEXT CLASSIFICATION




**Sentiment Analysis** is the application of natural language processing to systematically identify the emotional and subjective information in texts.

# POSSIBLE USE-CASE: TEXT


**Machine Translation** (e.g. google translate) Neural machine translation exploits neural networks to predict the likelihood of a sequence of words, typically modeling entire sentences in a single integrated model.

# APPLICATIONS OF DEEP LEARNING: SPEECH

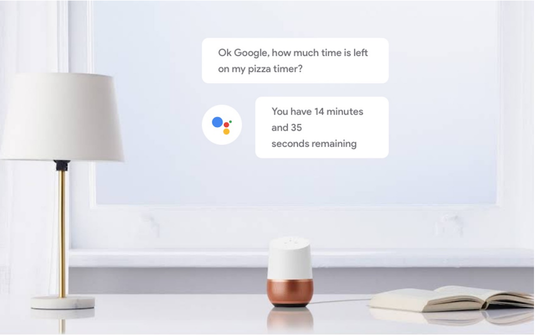



**Figure:** Example for speech recognition and generation (Google)

**Speech Recognition and Generation** (e.g. google assistant) Neural network extracts features from audio data for downstream tasks, e.g., to classify emotions in speech.